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Machine Learning for Robotics I

Machine Learning Labs Report 01

Navier Bayes classifier

Submitted By:
Tifrani Mohammed
ID: 8923873

Submitted To:
Prof. Stefano Rovetta

Lab 1: Naive Bayes Classifier for Weather Prediction

TIFRANI MOHAMMED ID:8923873
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***Abstract*—This report investigates the implementation of a Naive Bayes classifier using the Weather dataset with another larger dataset (the breast cancer). The classifier is designed to determine whether outdoor activities can be carried out based on various weather conditions. As part of the data preprocessing stage, categorical features were converted into numerical representations. The algorithm calculates prior probabilities and likelihoods for the categorical attributes, while Laplace smoothing is applied to prevent zero-probability problems. The experimental results demonstrate the classifier's effectiveness and its ability to perform reliably on small datasets containing sparse categories. Finally, the report discusses the main advantages and limitations of the proposed approach and identifies potential areas for future improvement. The "Think Further" section enhances the Naive Bayes classifier by implementing log-probabilities to eliminate numerical underflow and discretizing continuous Iris features via median thresholding, achieving a 73.33% classification accuracy**

I. Introduction

The Naive Bayes classifier is a supervised machine learning algorithm that uses probability theory to assign data samples to different classes. It is based on Bayes' theorem and estimates the probability of each possible class given the observed features. The algorithm assumes that the features are conditionally independent of one another when the class is known. Although this assumption is often unrealistic in real-world datasets, it greatly simplifies the calculations and makes the method computationally efficient. Naive Bayes is particularly useful for classification problems involving categorical or discrete data and can provide good results even when the available dataset is relatively small.

The goal of this lab is to:

1) Develop and implement a Naive Bayes classifier for a dataset containing categorical features t.
2) Improve the classifier's ability to handle previously unseen feature values by applying Laplace (additive) smoothing. .
3) Assess and compare the classifier's performance both with and without smoothing using a small weather dataset. .

II. Description of the Approach

A. Data-set Import

The weather data-set consists of 14 samples with 4 categorical features:

- **Outlook:** Sunny, Overcast, Rainy (3 levels).
- **Temperature:** Hot, Mild, Cool (3 levels).
- **Humidity:** High, Normal (2 levels).
- **Windy:** True, False (2 levels).
- **Play** The target variable indicates whether or not to play tennis (Yes/No, 2 levels).

B. Data Preprocessing

To prepare the data-set for the Naive Bayes classifier:

- Nominal String Format: Categorical variables were maintained as nominal strings (e.g., 'sunny', 'overcast', 'no-recurrence events') rather than being converted to integer codes, taking advantage of Python dictionaries to store strings directly as keys.
- The data set was divided into two subsets, with 70% of the data used for training and the remaining 30% reserved for testing.

C. Naive Bayes Classification

The Naive Bayes classifier is a probabilistic supervised learning algorithm based on Bayes' theorem. It is used to estimate the probability of a class given a set of observed features. Bayes' theorem is expressed as:

$$P(C \mid X) = \frac{P(X \mid C)P(C)}{P(X)} \tag{1}$$

where:

- $P(C \mid X)$ is the *posterior probability*, representing the probability that an observation belongs to class C given the feature set X.
- $P(X \mid C)$ is the *likelihood*, representing the probability of observing the feature set X given class C.
- $P(C)$ is the *prior probability* of class C.
- $P(X)$ is the *evidence*, representing the overall probability of observing the feature set X.

For a feature vector $X = (X_1, X_2, \ldots, X_d)$, the Naive Bayes classifier assumes that the features are conditionally independent given the class. Therefore, the likelihood can be factorized as:

$$P(X \mid C) = \prod_{i=1}^{d} P(X_i \mid C) \tag{2}$$

Consequently, since $P(X)$ is the same for all possible classes, the posterior probability can be expressed proportionally as:

$$P(C \mid X) \propto P(C) \prod_{i=1}^{d} P(X_i \mid C) \tag{3}$$

For categorical features, the conditional probability of a feature value given a class can be estimated from the training data as:

$$P(X_i = x_i \mid C) = \frac{N_{i,x_i,C}}{N_C} \tag{4}$$

where:

- $N_{i,x_i,C}$ is the number of observations in class C for which the feature X_i takes the value x_i.
- N_C is the total number of observations belonging to class C.

The prior probability of class C is calculated as:

$$P(C) = \frac{N_C}{N} \tag{5}$$

where N is the total number of observations in the dataset.

The predicted class is then obtained by selecting the class with the highest posterior probability:

$$\hat{C} = \arg\max_C \left[P(C) \prod_{i=1}^{d} P(X_i = x_i \mid C) \right] \tag{6}$$

D. Laplace Smoothing

To improve the robustness of the Naive Bayes classifier when dealing with previously unseen feature values, Laplace smoothing was applied. Without smoothing, a feature value that does not occur in the training set for a particular class would have a conditional probability of zero. Since the Naive Bayes likelihood is computed as a product of conditional probabilities, this would cause the entire likelihood to become zero and could lead to an incorrect classification.

Laplace smoothing modifies the estimation of the conditional probability as follows:

$$P(X_i = x_i \mid C) = \frac{N_{i,x_i,C} + \alpha}{N_C + \alpha V_i} \tag{7}$$

where:

- α is the smoothing parameter, typically set to $\alpha = 1$ for standard Laplace smoothing.
- V_i is the number of possible values that feature X_i can take.
- $N_{i,x_i,C}$ is the number of observations belonging to class C for which feature X_i has the value x_i.
- N_C is the total number of observations belonging to class C.

By adding α to the numerator and αV_i to the denominator, Laplace smoothing ensures that every possible feature value receives a non-zero probability. This allows the classifier to handle unseen feature values while maintaining a valid probability distribution.

III. Experiments and Implementation

A. Experimental Setup

- **Training Set:** 10 randomly samples from the data-set.
- **Test Set:** 4 remaining samples.
- **Evaluation Metric:** Error rate, calculated as:

$$\text{Error Rate} = \frac{N_{\text{incorrect}}}{N_{\text{test}}} \tag{8}$$

The classifier was tested under two scenarios:

- Without Laplace smoothing.
- With Laplace smoothing ($\alpha = 1$).

B. Results

Standard Naive Bayes Classifier classification results summarized in Table I:

TABLE I
NAIVE BAYES CLASSIFIER CLASSIFICATION RESULTS

Test Samples	Predicted Labels
yes	no
yes	yes
no	no
yes	yes
no	yes

$$\text{Error Rate} = 0.4$$

Standard Naive Bayes Classifier classification results with Laplace smoothing summarized in Table II:

TABLE II
NAIVE BAYES CLASSIFIER CLASSIFICATION RESULTS WITH LAPLACE SMOOTHING

Test Samples	Predicted Labels
yes	yes
yes	yes
no	no
yes	no
no	yes

$$\text{Error Rate} = 0.4$$

IV. Discussion

1) **Performance Without Laplace Smoothing (Weather Dataset):** In the initial experiment on the Weather dataset, the Naive Bayes classifier was evaluated without smoothing ($\alpha = 0$). The classifier achieved an accuracy of 60.00%, correctly predicting three out of five test instances. The remaining two test samples were misclassified, resulting in an error rate of 40.00%. This shows that while standard maximum likelihood estimation can yield reasonable predictions on simple test instances, raw frequency calculations remain vulnerable to zero-probability zero-out errors when encountering unobserved combinations.
2) **Performance With Laplace Smoothing (Weather Dataset):** Standard Laplace smoothing was introduced with $\alpha = 1.0$. The test accuracy remained unchanged at 60.00%. Although Laplace smoothing successfully prevents zero likelihoods and ensures robustness against unobserved levels, the updated posterior probabilities did not alter the final class assignments for this specific test split.

3) **Effect of Different Smoothing Parameters (Weather Dataset):** Further experiments were conducted using a stronger prior belief ($\alpha = 2.0$) and a weaker prior belief ($\alpha = 0.5$). Both configurations maintained an accuracy of 60.00%. This indicates that while varying α shifts the numerical likelihood values, the relative posterior ranking between classes remained stable across all tested smoothing weights for this small test set.
4) **Analysis of the Weather Classification Results:** For the test set, the actual ground-truth labels were ['yes', 'yes', 'no', 'yes', 'no'] while the model predicted ['no', 'yes', 'no', 'yes', 'yes']. The classifier correctly matched the second, third, and fourth observations, while misclassifying the first and fifth instances, yielding an accuracy of 60.00% ($3/5$ correct predictions).
5) **Evaluation on the UCI Breast Cancer Dataset:** When evaluated on the larger UCI Breast Cancer dataset, the raw Naive Bayes classifier ($\alpha = 0.0$) achieved an accuracy of 75.90%. During testing, three samples (rows 2, 13, and 18) were automatically identified as containing unseen attribute levels not present in the training set and were safely discarded. Applying Laplace smoothing ($\alpha = 1.0$) improved the classification accuracy to 78.31%. Testing alternative smoothing strengths ($\alpha = 2.0$ and $\alpha = 0.5$) yielded the same 78.31% accuracy, demonstrating that additive smoothing reliably prevents zero-probability failures while enhancing performance on larger categorical data.
6) **Continuous Feature Handling (Iris Dataset Experiment):** In the "Think Further" section, continuous features from the Iris dataset were discretized into binary representations using a median split threshold. Evaluating the binarized Iris dataset with Laplace smoothing ($\alpha = 1.0$) achieved a classification accuracy of 73.33%. This confirms that non-parametric discretization effectively allows categorical Naive Bayes implementations to process continuous inputs without requiring continuous probability density modeling.

V. Conclusion and Future Work

This lab successfully demonstrated the design, implementation, and empirical evaluation of a naive Categorical Bayes classifier. Using the small Weather dataset provided an accessible environment for verifying likelihood calculations and observing the behavior of frequency estimates. Evaluating the model on the larger UCI Breast Cancer dataset confirmed its scalability across higher dimensions, higher level counts, and missing attribute data.

The integration of Laplace (additive) smoothing proved crucial for addressing zero-probability issues caused by unobserved attribute levels, ensuring that unseen test patterns do not result in complete classification failures. In the Breast Cancer dataset, Laplace smoothing ($\alpha = 1.0$) increased performance from 75.90% to 78.31%. Furthermore, transforming probability products into log-sum calculations prevented numerical floating-point underflow. Finally, discretizing the Iris dataset using median splits confirmed that continuous attributes can be efficiently processed using categorical models.

Future work could focus on further extending and refining the model through:

1) **Gaussian Naive Bayes:** Extending the classifier to handle continuous features directly by calculating class-conditional means ($\mu_{c,j}$) and variances ($\sigma^2_{c,j}$) under a Gaussian probability density function.
2) **Dynamic Feature Discretization:** Exploring advanced binning strategies (such as multi-bin quantile splits or decision tree-based binning) rather than simple binary median splits.

References

[1] Machine Learning for Robotics I Course Slides in University of Genoa.